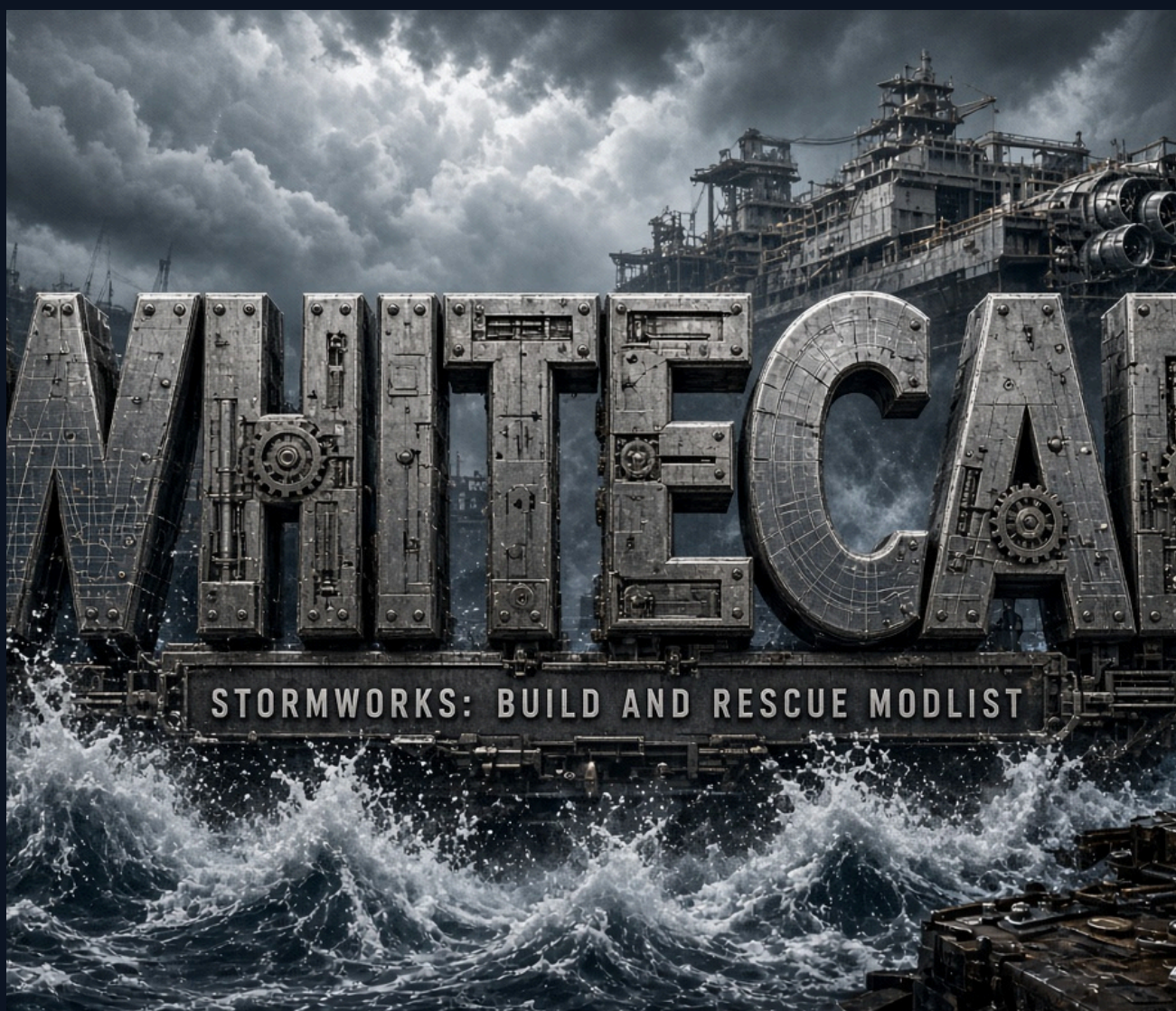




Whitecap

Stormworks: Build and Rescue v1.15.18

Modlist · Installation · Field Manual · Personas

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1 Installation

From a vanilla Stormworks v1.15.18 to a modded rescue service in four steps: game, mods, world, verify.

1.1 What you need

Item	Value
Base game	Stormworks: Build and Rescue v1.15.18 — Steam (current stable; Geometa is still actively developing the game)
DLC	All three, required: Search and Destroy (weapons & explosives), Industrial Frontier (industry), Space (space & mining) — the “Complete Collection” bundle covers them
Mod manager	The game’s official Mod Manager — Steam Workshop subscriptions + per-save “Enabled Mods” (asset modding since v1.13.0). No third-party tool needed.
Mods	The curated Workshop mods from chapter 2, installed in two layers
Sources	Steam Workshop · the game’s own Mod Manager · the Geometa modding wiki

Stormworks is a **living game**: Geometa ships several updates a year (v1.15.18 “The Airburst Update” landed June 2026). Steam keeps the game on the latest build automatically — there is no version selection to worry about, but it does mean mods must be checked against the current game version, because a game update can briefly break a mod until its author catches up.

All three DLCs are required for this modlist. The game’s current content — and the mods built on it — assume the full set: Search and Destroy for weapons and explosives (R.A.M.B.O. rebalances weapon components; several parts mods add military hardware), Industrial Frontier for mining and industry, and Space for rockets and orbital content. They install like the base game (one Steam purchase, no launcher, no options), and the **Complete Collection** bundle on the store page includes all three plus the base game in one discounted package.

Good to know

The two-layer structure exists for exactly this reason: **Layer 1** (QoL and graphics only) is your first save, and **Layer 2** (the full modded experience) is a second save you start once you know the ropes. Both are described in chapter 2, and both are created exactly the same way — the only difference is which mods you enable.

1.2 Step 1 — Install and launch the game

1. Install **Stormworks: Build and Rescue** from Steam.
2. Launch the game once and start a new game, so the game generates its save folders under %AppData%\Roaming\Stormworks\.
3. Quit and confirm the vanilla game runs before touching mods. If something is wrong, fix it first.

1.3 Step 2 — Subscribe to the mods

Every mod in this guide lives on the Steam Workshop — there is no separate download site and no external loader. The Workshop is also the **manager**: subscription, version tracking and updates are handled by Steam automatically.

1. Open the Steam Workshop for Stormworks: steamcommunity.com/app/573090/workshop.
2. Open each mod's page (every card in chapter 2 links its own page) and click **Subscribe**.
3. Layer 1 first — that is the save you start with.

Tip

Steam **Workshop Collections** let you group mods and subscribe to the whole group in one click — this is the avid modder's way to keep the two layers as two collections (“Whitecap Layer 1”, “Whitecap Layer 2”) and to switch between them cleanly.

1.4 Step 3 — Create the world and enable the mods

Stormworks decides which mods run **per save, at world creation**. The flow (documented by the R.A.M.B.O. author, and identical for every mod):

1. From the main menu click **New Game**.
2. Click **Enabled Mods**.
3. Switch to the **Workshop** tab.
4. Enable the mods for the layer you are starting (all of Layer 1 — or all of Layer 1 + Layer 2 once you have experience). Leave the rest off.
5. Click **Back**, set the rest of your world options (mode, difficulty — see chapter 4), and click **Confirm**.
6. Play a little, save, and quit.

Watch out

Mods are bound to the save at creation. If you later want a different mod set, **start a fresh save** with the new selection — do not try to bolt Layer 2 onto your Layer 1 save. (The advanced scene.xml path below exists, but it is exactly the kind of fiddly surgery a fresh save avoids.)

1.5 Step 4 — Verify the first launch

1. Launch the game and load your new save.
2. Confirm the mods are actually active — each mod's page describes what you should see (e.g. Opal's shader changes are immediately visible in the water and sky; Echo's workbenches are visibly bigger).
3. **If a mod is tagged with a game version older than the current build**, watch for issues — see “Updating and removing mods” below.

Watch out

R.A.M.B.O. (Layer 2) has one documented quirk: after launching the game, **load the modded save twice** — the first load does not always apply the modded physics (a game bug the author reports to Geometa). This is in the mod's own installation instructions.

1.6 Updating and removing mods

- **Update a mod**: nothing to do — Steam updates Workshop subscriptions automatically. If an update breaks something, check the mod's page for a fix before reporting it.

- **Remove a mod:** **Unsubscribe** on its Workshop page, then remove any vehicles that use its parts **before** loading the save without it — modded parts on a save without the mod can lock vehicles or corrupt the save.
- **Clean slate:** if you want every trace of a mod gone, Steam → right-click Stormworks → **Properties** → **Installed Files** → **Verify integrity of game files** resets the game’s own files.
- **The data/mods folder:** %AppData%\Roaming\Stormworks\data\mods holds local mod overrides. You generally never need to touch it — Workshop mods are managed in-game — but it is where mod files land if you download a mod manually, and removing a folder there is how you uninstall a manual mod.

Good to know

Adding a mod to an **existing** save is possible but unsupported and fiddly: it means editing scene.xml in the save folder (%AppData%\Roaming\Stormworks\saves\<save>) to reference the mod’s files. Create a temporary save with the mod enabled, copy the mod’s line from its scene.xml, and paste it into your main save’s scene.xml. Only do this if you cannot start a new save — it is the backup plan, not the flow.

1.7 Safety rules

- Back up your save folder: copy %AppData%\Roaming\Stormworks\saves\<save> somewhere safe before any big change. Saves are cheap to copy and expensive to lose.
- Read the **Requirements** and **Last updated** fields on every mod page before subscribing.
- Check the version tag on each mod (e.g. v1.15.18): a mod a few versions behind the game usually still works, but one that is years behind is a compatibility risk — this guide’s rejected list in chapter 2 is full of exactly those.
- Keep your Layer 1 and Layer 2 saves separate, and back up each before upgrading anything.
- The moment something breaks: verify the game files, then disable mods one at a time (new saves or the scene.xml path) to isolate the culprit.

2 The Modlist

What “Whitecap” actually consists of: Stormworks v1.15.18 plus a curated Steam Workshop modlist in two layers — and, just as importantly, what it deliberately leaves out.

2.1 The list at a glance

Property	Value
Base game	Stormworks: Build and Rescue v1.15.18 (current stable, “The Airburst Update”, 2026-06-24)
Required DLC	All three — Search and Destroy · Industrial Frontier · Space
Mod manager	Steam Workshop + the game’s built-in Mod Manager — per-save “Enabled Mods” (official asset modding since v1.13.0)
Layer 1 — “The Commissioning”	5 items — QoL and graphics only; the recommended first save
Layer 2 — “The Full Fleet”	11 more items — physics, parts, missions and detail, layered on top; start a fresh save once you have experience
Distribution	Steam Workshop — official pages only

Playstyle	The vanilla sim, deepened and modernized — nothing that makes it easier
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2.2 Design pillars

- **Deepen & modernize.** Stormworks is a living game whose physics and career are the point. These mods make its world feel deeper and more current — more realistic engineering, more parts to build with, more mission variety — without moving the goalposts.
- **Graphics, honestly.** One serious visual overhaul (Opal Shader) plus honest QoL. No fake “ultra” filters that only smear the image, and no second shader stacked on top just because it exists.
- **Details.** Physical touches a builder notices: sound you can hear (fog horns, thunder, turbo spool), geometry you can actually build with, an arctic you can walk on.
- **More AI, honestly.** Stormworks’ AI lives in its career mission generator — the game already spawns AI ships, aircraft, fire and rescue scenarios around you. This list deepens that system (Mango Cargo Overhaul generates far more varied AI missions; R.A.M.B.O. adds static NPCs on passenger seats). There is **no** maintained “smarter traffic” mod for Stormworks at guide time — the candidates are abandoned, and pretending otherwise would be fiction.
- **No overpowered, no easier.** Nothing that raises money, payloads, or success rates; nothing that erases a mechanic the game teaches you by building. Mods that do exactly that are listed at the end of this chapter with reasons.

2.3 The mod manager: Steam Workshop + the in-game Mod Manager

Stormworks has **official asset modding** since v1.13.0, and for an avid modder it is the only manager this guide uses:

- It is the ecosystem itself — every maintained mod is a Workshop item, so one manager covers everything: subscribe, auto-update, unsubscribe.
- It is **per save**: the in-game Mod Manager (“Enabled Mods” at New Game) lets you choose exactly which mods a world runs. That is what makes the two-layer structure possible on one installation.
- Steam **Workshop Collections** group mods for one-click subscription — keep a “Whitecap Layer 1” and a “Whitecap Layer 2” collection and switching is trivial.
- It needs no external launchers, no file juggling, and no third-party tools.

Good to know

Legacy external loaders — **Stormloader** and friends — predate official modding and are obsolete. If an old mod page asks you to install a loader, treat it as a legacy-mod red flag.

2.4 Layer 1 — “The Commissioning” (QoL & graphics)

The starting save. Nothing here changes how the game plays — it makes it look better, builds easier, and removes only genuine annoyances.

2.4.1 Graphics

Opal Shader 2.6

Visual overhaul · by Dheix

The community-standard shader overhaul for Stormworks: remade water and clouds, better atmosphere and sky, grass and snow, world curvature, net sails and fire effects. It is the single biggest modernization the game's

look can get, and the ecosystem's de-facto default. Not compatible with any other shader or particle mod — which is fine, because it replaces the need for them.

Where: [workshop: 3393471780](#)

Install: Subscribe and enable at world creation. For grass to work, the Grass Module below must be enabled too.

Risk: GPU-heavy — lower quality settings on weak PCs. Do NOT combine with Project Reshade, Sea Lanes, SunRays or any other shader mod.

Opal Shader - Grass Module

Required companion · by Dheix

The grass meshes Opal needs: adds the models for the grass rendering to the vanilla tiles. Without it, Opal's grass simply does not appear.

Where: [workshop: 3393475544](#)

Install: Subscribe and enable alongside Opal Shader.

Risk: Low. Vanilla tiles only — does not work with modded tilesets.

Opal Shader - Config Module

Optional customization · by Dheix

A small config file that unlocks Opal's settings — snow, custom fog color, world curvature, grass, net sail, rain reflections, new fire. Edit `\_opal\_config.glslh` in any text editor and enable the module at world creation to apply your choices.

Where: [workshop: 3393475869](#)

Install: Subscribe and enable at world creation, then edit `\_opal\_config.glslh` to taste.

Risk: Low — a text file. Defaults are sensible if you never touch it.

2.4.2 Quality of life

Echo's Bigger Workbenches v11

QoL · by Echoless

The definitive workbench mod, rebuilt from the current game files: almost every workbench in the game gets a significantly larger build area, including extended train workbenches. You stop fighting the build envelope and start building.

Where: [workshop: 3357835209](#)

Install: Subscribe and enable at world creation. It is a clean-slate rebuild — do not stack older workbench mods on top.

Risk: Low. Environment-only changes (build zones).

Better Hitboxes

QoL · block precision

Removes invisible hitboxes from a growing list of blocks (push buttons, toggles, keypads, instrument panels, dials, compasses...). Logic nodes stop nesting inside each other and blocks stop colliding where they shouldn't — docking and panel layouts behave the way they look.

Where: [workshop: 3359500787](#)

Install: Subscribe and enable at world creation.

Risk: Low-to-medium. It rewrites block definitions — keep a save backup before updating the mod (the author's own advice). Stays in Layer 1: it touches stock component definitions, so keep it out of the R.A.M.B.O. save unless you enjoy debugging overlaps.

2.5 Layer 2 — “The Full Fleet” (the full modded experience)

Start a fresh save with everything Layer 1 has plus these. This is the layer for players who have built, flown and floated a few things and want the sim to bite back — honestly, and with more to do.

2.5.1 The physics centerpiece

R.A.M.B.O. — Realistic Aerodynamics, Mass & Buoyancy Overhaul Physics overhaul · by Thales · the heart of Layer 2

The mod that makes Stormworks engineer honestly: realistic hydrodynamics and aerodynamics, component mass rebalanced toward powerplants/fuel/payload, nerfed stock propulsion with realistic thrust-to-weight, shape now matters for drag in air and water, sealed interiors and ballast become real design tools, and it adds paintable surfaces, static NPCs on seats and sound changes. Stock components only — no new parts — so vehicles stay two-way compatible with unmodded saves. Fully documented in its Steam Guide (id 3736924997).

Where: workshop: 3737924863 · guide: 3736924997

Install: Subscribe, enable at world creation — then remember the documented quirk: load the save TWICE after every game launch, or the modded physics won't apply.

Risk: Medium by design: it makes vehicles heavier, slower and more demanding. Incompatible with mods that change game constants or stock component definitions (hence Better Hitboxes stays in Layer 1).

Good to know

R.A.M.B.O. is the reason the two layers exist. It does not make the game easier — it makes the physics real enough that a badly designed boat sinks and an underpowered plane won't climb. That is the “full modded experience” this layer is for.

2.5.2 Parts & building depth

Frustum aka Square Pyramids

Parts · recommended by R.A.M.B.O.'s author

33 frustum (square-pyramid) blocks with individually paintable faces — the geometry base game never shipped. Pairs naturally with R.A.M.B.O.'s paintable-surface overhaul.

Where: workshop: 3657682402

Install: Subscribe and enable at world creation.

Risk: Low. Additive component mod.

Thales' Aviation Parts and Block Overhaul — Updated

Parts · by Thales

250+ aviation and block parts (wings, control surfaces, fuselage pieces, slopes) that fill the shape gaps for serious aircraft builders. The maintained successor to the original — vehicles built with the legacy version load seamlessly into this one.

Where: workshop: 3535295022

Install: Subscribe and enable at world creation. Do not also enable the legacy version (3357808533).

Risk: Low. Additive parts mod; same author and philosophy as R.A.M.B.O.

Electric Parts Expanded

Parts

New electric parts that fill gaps for small builds — drones, compact vehicles — styled and balanced to match the base game.

Where: workshop: 3357494697

Install: Subscribe and enable at world creation.

Risk: Low. Additive component mod.

Mechanical Parts Expanded

Parts

New mechanical parts in the same spirit — base-game style and balance, filling the gaps for transmission and machinery builds.

Where: [workshop: 3538324086](#)

Install: Subscribe and enable at world creation.

Risk: Low. Additive component mod.

Structural Parts

Parts

New structural shapes that round out the hull and frame vocabulary for cleaner, more detailed builds.

Where: [workshop: 3541463211](#)

Install: Subscribe and enable at world creation.

Risk: Low. Additive component mod.

2.5.3 Missions & world

Mango Cargo Overhaul

Missions & economy

Overhauls the cargo delivery system: kiosks generate cargo packs by port type and biome, with options for locked routes, passengers (ferry and airport runs), trailers and standardized containers. The default container shuffle becomes a real logistics career — more missions, more AI traffic, more reason to build proper freighters.

Where: [workshop: 3635338496](#)

Install: Subscribe and enable at world creation (spawns its kiosks in the world).

Risk: Low-to-medium. Actively maintained; a few edge cases (custom kiosks) remain in-progress per the author.

ARCTIC+ | Arctic Ice Pack

World detail

A large, walkable ice sheet on the arctic sea — drive vehicles across it, land aircraft on top, dive underneath with a submarine. The arctic stops being a dead end and becomes a place.

Where: [workshop: 3703818116](#)

Install: Subscribe and enable at world creation — the ice appears in new saves only.

Risk: Low. Save-scoped environment change.

2.5.4 Sound & detail

Realistic Foghorn / Whistle

Sound detail

A realistic foghorn for your vessels — does not replace the vanilla horn, adds alongside it, with pitchable whistles. Small, physical, and exactly the kind of detail a mariner notices in a fog bank.

Where: [workshop: 3367942963](#)

Install: Subscribe and enable at world creation.

Risk: Low. Sound-only, non-replacing.

Better Thunder

Sound detail

Replaces the thunder sounds with deeper, more atmospheric ones. Stormworks storms are a big part of the drama; this makes them sound like it.

Where: [workshop: 3523467712](#)

Install: Subscribe and enable at world creation.

Risk: Low. Sound-only.

Realistic Turbocharger

Engine depth

A realistic turbocharger with accurate spooling behaviour and sounds — pressure builds with exhaust volume, different engines spool differently. Engine design gets one more real variable, and turbos stop being free power.

Where: [workshop: 3521082511](#)

Install: Subscribe and enable at world creation.

Risk: Low-to-medium. Changes engine behaviour — allow exhaust volume to fill before judging the spool (author's note).

2.6 What we deliberately left out

Every mod below is real and some are very popular — and each one fails a pillar. The reasons are recorded in STATUS.md as well.

- **Project Reshade, Sea Lanes Shader, SunRays** — other shader overhauls. Opal is the pillar; the rest either conflict with it explicitly (“not compatible with Opal”, “does not work with Opal”) or ship documented visual bugs (Sea Lanes’ tsunami/whirlpool rendering). One good shader, not three fighting ones.
- **DEEPER OCEANS MOD** — a dramatic whole-world seabed overhaul (everything 3,000 m deep), but unmaintained since early 2025 through four game versions. The world-changing mods that survive are the ones that keep up (see ARCTIC+).
- **North Sawyer Overhaul** — the most-subscribed world overhaul, and exactly the wrong direction for this list: it rewrites major locations, is unmaintained since v1.13.1, and its philosophy is replacement rather than deepening. When world mods drift, they break saves — this one is drift.
- **ilpottis Decoration Mod** — 289 lovely parts, but the author itself calls the current release “a direct port” with a full rewrite pending, unmaintained since v1.13.0. The Parts Expanded trio covers the pillar on current game files.
- **Improved Default Mission** — more mission variety, but last updated 2022, predating the official mod system. Mango Cargo Overhaul covers mission depth on current files.
- **QOL microcontroller series** (smaller logic chips) — pure convenience that removes space-management friction from the engineering puzzle. Shrinking your circuits is part of the game.
- **Workbench mods other than Echo’s** (Better Workbenches, Bigger Workbenches, lowPoly Workbench) — superseded by Echo’s v11, which is rebuilt from the current game files and is the most-subscribed.
- **“All-in-one” modpacks** (e.g. SB mod) — uncurated bundles that mix unrelated mods, several of which this guide rejects on principle. Curating is the point of a modlist.
- **Stormloader and legacy external loaders** — obsolete since official Workshop modding.

Good to know

The line we draw: **friction you can feel is the game**. Detail mods, honest physics, and more to build with deepen it; anything that quietly removes a mechanic, a limit, or a source of tension is out — even when it is popular.

2.7 Download sources

- Steam Workshop — Stormworks hub: steamcommunity.com/app/573090/workshop
- Geometa modding wiki: geometa.co.uk/wiki/stormworks

Every mod card above links its own Workshop page — check the **Requirements** and **Last updated** fields there before installing; the game updates several times a year and versions move faster than any printed guide.

3 How to Play

A field manual for the modded Stormworks: what the game is, how the career loop works, how to design vehicles that float, fly and function — and what the mods change once you are in.

3.1 What the game actually is

Stormworks is a **block-based physics sandbox with a rescue career on top**. You build vehicles — boats, planes, helicopters, submarines, trucks — from voxel blocks, wire them with electrics, fuel and logic, and then use them to run a search-and-rescue service in a 15,000 km² ocean world with islands, an arctic, dynamic weather and day/night cycles. Physics are the game: displacement-based buoyancy, modular engines, drag, damage, fires.

There are four entry points:

- **Career** — the full sim: you start with a budget, buy vehicles and bases, and complete missions to fund growth.
- **Classic Career** — the original career structure: pre-placed missions around the world, no economy creep.
- **Creative** — unlimited building; the sandbox for designing without cost or survival constraints.
- **Custom** — a career with a huge settings panel: you toggle **survival options** (fuel consumption, engine overheat, electricity, vehicle damage, fires, weather, player health...) and the economy. This is where “game creation” actually happens — see chapter 4.

Tip

Most players’ first 20 hours are Creative — learning to build. The career is the goal; the sandbox is the school.

3.2 The rescue loop

Career gameplay is a loop: **take mission** → **pick the right vehicle** → **go out** → **complete it** → **get paid** → **build better**. Missions generate around the world and tell you what you need:

- **Sea rescue** — survivors in the water or on a sinking vessel; you need a fast, seaworthy boat (and a winch or deck space).
- **Fire** — burning vehicles or structures; you need a water tank, pumps and hoses.
- **Cargo & logistics** — move containers, fluids or passengers between ports; you need carrying capacity and range.
- **Exploration** — reach coordinates and investigate; range and navigation matter.
- **Salvage / deep-sea** — recover wrecks or work underwater; you need winches, divers or a submarine.

Every mission pays, and the payouts fund the next vehicle. Vehicles are spawned at **workbenches** — the build/spawn pads at each base — and bases are bought and upgraded as you expand. Money is the career’s only real pressure: it decides what you can build, not what you can do.

Good to know

In Layer 1 this is exactly the vanilla career, prettier and with more build room. In Layer 2, Mango Cargo Overhaul turns the cargo branch into a proper logistics game (kiosks, port types, ferry routes), R.A.M.B.O. makes every vehicle you design a real engineering problem, and ARCTIC+ gives the map's far edge somewhere to go.

3.3 Your first 20 minutes

A guided first sortie: from the menu to your first completed mission. The exact keys are the game's defaults — the Settings screen shows every binding, and vehicle dashboards often add their own buttons (power, starter, reverse).

Control (default)	What it does
W / A / S / D	Move on foot; from a seat: throttle / brake / steer (W and S control speed, A and D turn)
Mouse	Look; scroll to zoom the camera
E	Interact — enter/exit seats, pick up items, use switches, grab survivors
M	Open the world map with mission markers
Shift (hold)	Sprint on foot
Start engine	Per-vehicle: a dashboard starter button, or the game's start-engine key — check Settings

3.3.1 Minutes 0–2 — start the right world

1. **New Game** → **Career** (this guide assumes the Layer 1 mods from chapter 1 are enabled — see chapter 1 for the “Enabled Mods” flow).
2. Confirm the world and you spawn at the starter base on the main island (Sawyer Island): a dock, a small workbench, and your first vessel — the game's **Starter Boat** preset.
3. Walk around for a minute: find the dock, the workbench pad, and the fuel pump. You will use all three constantly.

3.3.2 Minutes 2–5 — read the boat and the map

1. Board the starter boat (interact) and sit in the pilot seat.
2. Read the dashboard: a **master power** switch, an **engine starter** button, a **reverse** toggle, and the gauges (fuel, RPM, speed). Find the **hose anchor** — that is where fuel docks connect.
3. Open the map (M). A **mission marker** should already be there — the game spawns a tutorial rescue shortly after you start. Note the transponder locator on the dash: it points you at mission signals directly.

Good to know

The starter boat has a gyro stabilizer — it fights rolls for you. That is a hint about how most good small boats are built: stable first, fast second.

3.3.3 Minutes 5–10 — first sortie

1. Master power ON, then hold the throttle slightly open (W) **while** pressing the engine starter — a cold engine usually needs a little throttle to catch.
2. Ease off the dock, then A/D to steer toward the marker. Keep the throttle moderate — the starter boat is not a racer, and rough water will punish sharp turns.

3. If the engine dies, check fuel and re-start: throttle open a touch, starter again. If it dies instantly every time, you have no fuel — see refuelling below.

3.3.4 Minutes 10–15 — the rescue

1. Slow down as you approach the marker. Use the map and the transponder to find the survivor(s) — often a small raft or a person in the water.
2. Pull alongside gently, get close, and interact to pick them up / board them. Stow them on deck or in the cabin.
3. Head back to the base — the return trip is the part where boats flip, so keep it steady.

3.3.5 Minutes 15–20 — complete, save, and look around

1. Dock at the base. Completing the mission pays out — watch the money counter.
2. Open the **workbench** and enter the build editor. Spend the last five minutes **taking the starter boat apart and putting it back together** — deleting the engine, seeing how the propeller, rudder, fuel tank and wiring connect. This is the real first lesson; the next section is built on it.
3. Save the game (exit or the save option — the game autosaves on exit, but save explicitly after milestones).

Watch out

Do not take the starter boat into the open ocean in bad weather, and do not ignore the fuel gauge — the refuel lesson is: dock at a fuel pump, and connect the hose from the pump to the boat's hose anchor. Running dry mid-rescue is the classic first death.

Tip

If missions come too fast (or too slow) for comfort, the world settings let you adjust mission spawn frequency — a 20–30 minute pace is a calm start. Chapter 4 shows how the same settings become your persona's difficulty dial.

3.4 Engineering: building stuff that works

3.4.1 Blocks, seams and watertightness

Everything is built from **blocks** (voxels): cubes, wedges, pyramids, and the specialized parts (engines, tanks, instruments, logic). Two facts rule water:

- A hull only floats if it is **watertight** — every seam where blocks meet must be sealed. The game marks leaks: open seams show as gaps between blocks, and a leaking hull floods.
- **Displacement** does the floating: the volume of hull below the waterline pushes water aside, and that displaced water's weight is the buoyancy. Bigger submerged volume = more buoyancy. A dense, heavy vehicle sits lower in the water — its **draft** is deeper.

Practical rules: build the hull as a closed box first, place internal tanks and machinery, then **test before detailing**. If it sinks at the dock, the problem is almost always a seam, an open block face, or a hatch left unsealed.

3.4.2 Engines and power

The standard power train: **engine** → **generator** → **battery** → **consumers**.

- **Modular engines** are built from parts — cylinders, crankshaft, pistons, exhaust manifolds — and need **fuel, air, cooling and exhaust** to run. They produce mechanical power at an RPM (measured in **RPS**, rotations per second).
- Mechanical power turns generators, propellers, rotors, wheels and pumps. Electric power (batteries + generators) runs instruments, logic, lights and small motors.

- **Fuel systems:** tanks, pumps and pipes. A classic failure: the pump is on the wrong side of the tank, or the tank is part of a non-watertight volume.

Watch out

The most common “why won’t it start” causes, in order: no fuel (or fuel pump off), dead battery, engine not getting air or coolant, exhaust blocked, or a broken throttle/logic connection. Diagnose in that order.

3.4.3 Controls, rotors and lift

- **Boats:** throttle to engines/propellers, rudders or pivoting outboard drives for steering. Trim matters — a bow-heavy boat plows.
- **Helicopters:** main rotor for lift, tail rotor (or ducted fan) against torque, cyclic/pitch control via rotator servos or control surfaces. Helicopters need constant power and are unforgiving of weight.
- **Planes:** wings generate lift from forward speed; control surfaces (ailerons, elevators, rudder) steer. Underpowered = won’t climb; overpowered = unstable.
- **Cars/trucks:** wheels + engine + brakes; steering via the steering mechanism, pivot wheels, or differentials for heavy rigs.

3.4.4 Logic and microcontrollers

Stormworks has a full logic system: **sensors** (GPS, gyro, compass, altitude, distance, fluid level...), **logic gates**, and **microcontrollers** — programmable chips with LUA scripting. With microcontrollers you can autopilot a ship, auto-level a helicopter, manage a power grid, or make a fuel gauge that warns you at 20%. The basics: every sensor outputs a **composite** signal (multiple channels on one cable); every instrument displays one. The classic first project: a GPS-driven autopilot that holds a course between two waypoints. It is programming, but the pieces are small.

Tip

Start with the game’s included example microcontrollers and the in-game tutorials — then steal from your own workshop subscriptions. Half of learning Stormworks is taking someone else’s working boat apart.

3.4.5 Building, step by step

The **rule of three** covers every first build: keep the engine low (stability), make the hull/body fully enclosed (watertightness or sealed volume), and keep fuel, air and exhaust lines separate and connected. Every build below follows it.

3.4.5.1 The first boat (your first build, every time)

A 10–15 block long, 4–5 block wide rescue boat — nothing more.

1. **Hull:** build a closed box of blocks with wedges at the bow and a flat stern. Inside the box, nothing touches the outer skin — that is your volume.
2. **Seal it:** walk the hull and check every seam. A leak here sinks the whole career start.
3. **Engine:** place a small engine low and roughly centered — low mass = stable boat. Add a **generator** beside it and a **battery** somewhere dry.
4. **Drive:** a **propeller** at the stern, waterline-high, driven by the engine; a **rudder** directly behind it for steering.
5. **Fuel:** a **fuel tank** connected by pipes to a **fuel pump**, then to the engine. Add a **hose anchor** on deck for refuelling at docks.

6. **Air & exhaust:** an intake pipe to the surface and an exhaust pipe out the stern — blocked pipes stall engines.
7. **Seat & wiring:** a driver seat. Wire: seat **throttle** → engine throttle node; seat **steering** → rudder (through a pivot or steering block). Add a push-button wired to the engine's start node if you want a starter button.
8. **Test:** spawn it, check the waterline (it should sit with a modest draft), motor around the dock, then go fast in a straight line and turn hard — if it leans dangerously, add ballast low or widen the hull.

Watch out

The classic first-boat failures, in order: an unsealed seam (it sinks at the dock), the fuel pump wired backwards (it never reaches the engine), and a propeller above the waterline (lots of noise, no thrust).

3.4.5.2 The helicopter (once the boat floats)

1. **Core:** a compact fuselage box, a **driver/helo seat** in the middle, engine + generator + battery low in the frame.
2. **Lift:** a **main rotor** on top, driven by the engine; a **tail rotor** at the rear — without it the fuselage spins. Wire the tail rotor's throttle to keep it fighting the torque.
3. **Control:** the seat's **cyclic** (pitch/roll) and **collective** (throttle) outputs to the rotor's control nodes, or use rotators/control surfaces on the rotor head for pitch.
4. **Weight:** balance the center of mass under the rotor — a tail-heavy helicopter pitches back and crashes. Test in short hops, not full climbs.
5. **Automate later:** a gyro microcontroller that trims cyclic output is the difference between “it flies” and “it flies itself”.

3.4.5.3 The plane (fixed-wing)

1. **Fuselage:** a long closed box, seat in the middle, engine + fuel low in the nose or belly.
2. **Lift:** a pair of **wings** at the center of mass; **control surfaces** (ailerons on the wings, elevator on the tail, rudder on the fin).
3. **Propulsion:** a **propeller** at the nose (or two on the wings); thrust must exceed total drag or it will not climb.
4. **Trim:** the elevator should hold level flight with neutral stick — if the nose drops, move the center of mass forward of the wing center, and vice versa.
5. **Test:** taxi fast on land or water first; only climb once it reaches takeoff speed in a straight line. Landings are just slow glides — cut throttle, hold the nose up, touch down gently.

3.4.5.4 The car or truck

1. **Chassis:** a low flat box, 4+ **wheels** (small for a buggy, medium for a truck), engine + fuel in the middle.
2. **Drive:** wheels on the engine axle (direct) or through a gearbox for torque; **brakes** on at least the front axle — a Stormworks truck without brakes is a one-way trip.
3. **Steering:** a steering block or pivoting front wheels wired to the seat's steering output; keep the turning radius honest for the wheelbase.
4. **Weight:** keep the center of mass between the axles and low — a tall, heavy build rolls on the first turn.

Good to know

Every one of these builds changes under R.A.M.B.O.: mass is heavier, thrust is weaker, and shape decides drag. The rules stay the same — sealed volume, low engine, balanced mass, honest power — but the **tolerances** get real. Build the Layer 2 fleet deliberately; that is the point.

3.5 Engineering under R.A.M.B.O. (Layer 2)

R.A.M.B.O. changes the physics **philosophy**, not the rules of play — the documented changes that matter when you design:

- **Mass is real.** Components weigh what they should: engines, fuel, weapons and seats are significantly heavier; basic blocks are lighter and standardized (0.25 per voxel). What you carry now determines how your vehicle behaves.
- **Thrust is nerfed.** Propellers and engines produce less relative to weight — a realistic thrust-to-weight. The stock game's "slap an engine on it" shortcut stops working.
- **Shape matters.** Water drag is lower overall, but hull shape — especially the **rear** — now decides it. Air drag is dominated by the **front** shape; streamlined builds genuinely fly faster.
- **Buoyancy is fixed and consistent** (0.5 mass/L, with enclosed fluids at double density). A sealed hull floats more naturally; ballast and weight blocks become real tools for stability and trim.
- **Lift is proportional to wing size.** Wings work like wings; control surfaces have realistic deflection limits.

Watch out

R.A.M.B.O. is a fresh-design mod: vehicles built in stock will generally still work, but poorly — under-powered and overbuoyant. Build Layer 2 vehicles **for** Layer 2. And remember: load the save twice after each launch or the modded physics won't apply.

3.6 Basic terms

Term	Meaning
Block / voxel	The grid unit everything is built from; hulls, decks and structures are block assemblies.
Seam	The edge where two blocks meet. A watertight hull has no open seams below the waterline.
Displacement	The volume of water a hull pushes aside; its weight equals the buoyant force.
Draft	How deep a vessel sits in the water — the distance from waterline to keel.
Ballast	Added weight (usually pumped fluid) used to change trim or stability.
RPS	Rotations per second — the game's unit for engine and rotor speed.
Workbench	The build/spawn pad where vehicles are constructed and launched; bigger = bigger builds (Echo's mod).
Mission	A generated scenario (rescue, fire, cargo, exploration) with a payout.
Waypoint	A navigational target coordinate used by autopilots and maps.
Microcontroller	A programmable logic chip; sensors and gates wired in, LUA code inside.
Composite	A multi-channel signal carried on one cable — how complex data moves between parts.
Prefab	A pre-built sub-assembly (engines, cockpits) saved for reuse.
Mod / Addon	Workshop items: Mods change game files (shaders, physics, parts); addons/vehicles are content you spawn.

Survival options	Custom-mode toggles: fuel, overheat, damage, fire, weather, player health.
XML edit	Editing a vehicle/save file by hand to change values the editor doesn't expose — the power user's last resort.

3.7 The modded experience

What the two layers change once you are playing:

- **Opal Shader** — the sea, sky, weather and lighting are simply better. Fog banks, storms and night rescues stop being grey mush. If frames drop, lower the game's graphics quality first; the config module can disable specific effects (snow, grass, rain reflections).
- **Echo's Bigger Workbenches** — you will rarely hit the build envelope again; large ships and aircraft are buildable without tricks.
- **Better Hitboxes** — buttons and instruments on panels stop blocking each other; docking and dense control panels behave.
- **R.A.M.B.O.** — everything in "Engineering under R.A.M.B.O." above. The same world, but design choices now have consequences.
- **Parts expansions** (Frustum, Aviation, Electric/Mechanical/Structural) — new geometry and components in the editor; search their names in the parts list.
- **Mango Cargo Overhaul** — cargo kiosks appear at ports; deliveries come in types (containers, passengers, trailers), routes and port-locked contracts. The freight career is now a real logistics puzzle.
- **ARCTIC+** — the arctic sea is a walkable ice shelf: drive, land, and dive under it.
- **Sound mods** (foghorn, thunder, turbo) — listen: the world and the machinery sound like themselves.

3.8 Saves, backups & troubleshooting

- **Saving:** the game autosaves and saves on exit; keep manual copies of your save folder (%AppData%\Roaming\Stormworks\saves\<save>) before big changes.
- **Mod problems:** check the mod's **Last updated** date first — a game update is the usual culprit. Then check the mod's discussions; then disable mods one at a time to isolate.
- **R.A.M.B.O. physics not applying:** load the save twice (documented quirk). If components show modded features but physics feel stock, this is why.
- **Vehicles break after a mod update:** remove modded parts from vehicles **before** unsubscribing the mod — saves with orphaned mod parts can lock up.
- **Crashes or weirdness:** Steam → Verify integrity of game files resets the game; for manual mods, clear %AppData%\Roaming\Stormworks\data\mods.

3.9 How to get good (a checklist)

- Build in Creative first; test every vehicle at the dock before committing to it in career.
- Keep the first vehicles small and simple — a good 10-block boat beats an ambitious wreck.
- Seal the hull before decorating. Water finds every seam.
- Balance: check the center of mass vs. buoyancy before adding the superstructure.
- Learn one microcontroller pattern (autopilot or power management) early; it pays for itself.
- When a mission fails, the vehicle is usually the problem — that is the game working as intended.
- In Layer 2, expect to fail more and learn more. R.A.M.B.O. rewards the careful engineer, exactly like the sea does.

4 Personas

Who runs the service? Five ways to play the same Stormworks — each with a background, the world-creation choices that make it yours, and the mods that fit.

4.1 Before you start: how “game creation” works here

Stormworks has no character creator — no name, no face, no skill points. Your identity is what you **decide**, and the game gives you real levers:

- **The world.** At New Game you pick the mode — Career, Classic Career, Creative, or Custom — and in Custom you choose the **survival options**: fuel consumption, engine overheat, electricity draw, vehicle damage, fire spread, weather severity, player health. Those toggles are the difficulty dial, and they are the real “character sheet”.
- **The rules.** Every persona below sets its own house rules: what you may build first, what you must earn before you expand, what counts as winning. Difficulty comes from the rules you keep, not from a menu.
- **The mods.** The layer you start with is a persona choice too: Layer 1 for the first run, Layer 2 (fresh save) once you have sea legs. Each persona leans on the mods that matter to its playstyle.

Good to know

All five personas are playable on the same installation — personas differ in world settings, goals and self-imposed rules, not in the mods available. Start a separate save per persona and keep one backup per save.

4.2 1. The Mariner

The Mariner

Difficulty: medium · Focus: the sea

Background. You ran a coastal fishing boat for fifteen years and know exactly what a hull can and cannot take. The rescue service is a natural next step: a small starter boat, a dock, and a promise to answer every distress call within reach. The fleet grows hull by hull — a fast RHIB, a sturdy offshore vessel, a fireboat when the fires start.

Game creation choices

- **World:** Career (or Custom with survival options on: fuel consumption, engine overheat, vehicle damage, weather — the sea should matter).
- **First build:** a small, simple rescue boat — fast, stable, with a winch and deck space for survivors. No helicopters until the boats are proven.
- **House rules:** every mission answered with the right-sized vessel; no vehicle bought before the previous one has earned its keep; keep one boat in reserve at all times.
- **Goal:** a three-vessel fleet covering sea rescue, fire and cargo — every distress call within your area answered, no survivors lost.
- **Mods that shine:** Opal Shader (storms at sea finally look like storms), R.A.M.B.O. (seaworthiness becomes a design discipline), Echo’s Bigger Workbenches, Realistic Foghorn, Better Thunder.

4.3 2. The Aviator

The Aviator

Difficulty: medium · Focus: the sky

Background. You spent a decade flying rotary-wing SAR — hoists out of the surf, winching off cliff faces, long night transits in weather that grounds everyone else. Stormworks is your chance to build the perfect rescue helicopter, then the fixed-wing long-range platform that makes the whole map reachable.

Game creation choices

- **World:** Career with survival options on (fuel matters, overheat matters, weather matters). Range and endurance are the whole game.
- **First build:** a simple two-rotor or single-rotor helicopter — collective/pitch controls, tail rotor, winch. Master the hover before adding speed.
- **House rules:** every aircraft must be able to do an out-and-back sortie to its mission area on internal fuel; no passenger seat without a rescue role; night operations are mandatory, not optional.
- **Goal:** an air wing — SAR helicopter, cargo plane, and a long-range amphibian — covering the entire map including the arctic.
- **Mods that shine:** Thales' Aviation Parts (the shapes you need), R.A.M.B.O. (lift and thrust are now honest physics), Opal Shader, Better Thunder, Realistic Turbocharger.

4.4 3. The Industrialist

The Industrialist

Difficulty: medium · Focus: the economy

Background. You never wanted the heroics — you wanted the harbor. A shipping company that starts with one rusty coastal freighter and grows into the service every other persona calls for help: containers, tankers, ferries, trains. Money is the score, and logistics is the game.

Game creation choices

- **World:** Career with Mango Cargo Overhaul at the center of it (Layer 2). Economy on; survival options light (this is a business sim, not a survival sim).
- **First build:** a small cargo boat that can carry one container or a passenger run — earn, then scale.
- **House rules:** every new vessel must pay for itself within ten missions; no military or rescue hardware — if it does not move cargo, it does not get built; terminals, not heroics.
- **Goal:** a freight empire — container routes across the map, a ferry line, a train network, and the arctic trade route open.
- **Mods that shine:** Mango Cargo Overhaul (the heart of this persona), ARCTIC+ (ice-shelf trade), Structural Parts and the parts expansions, Echo's Bigger Workbenches, Opal Shader.

4.5 4. The Engineer

The Engineer

Difficulty: variable · Focus: the build

Background. You are not a rescue pilot — you are the designer everyone else flies. The workshop is your workshop: microcontrollers that hold a hover, hulls that don't leak, XML-edited geometry that looks hand-finished. You build in Creative, test in Custom with everything on, and publish what works to the Workshop.

Game creation choices

- **World:** Creative for design; a Custom test world with **all** survival options on — if it survives that, it survives the career.
- **First build:** anything, deliberately — the point is the process. A leak-proof hull with a microcontroller-managed power grid is the rite of passage.

- **House rules:** no workshop subscriptions for your own builds until you have built the equivalent yourself once; every published creation must run unmodded or state its mod requirements clearly; document the build.
- **Goal:** a portfolio — a boat, a helicopter, a plane and a submarine, each better than the last, at least one published to the Workshop.
- **Mods that shine:** everything — but especially R.A.M.B.O. (design against real physics), Frustum, Aviation Parts, the parts expansions, and Opal Shader (the showcase needs the lighting).

4.6 5. The Salvage Diver

The Salvage Diver

Difficulty: hard · Focus: the deep

Background. You worked saturation salvage rigs in a previous life and you have never lost the itch. The career is a cover story — what you actually want is the map's deep corners: wrecks, the arctic shelf, the places missions don't send anyone else. Every expedition needs a dive boat, a winch, and eventually a submarine that comes back.

Game creation choices

- **World:** Custom with survival options **hard:** fuel, overheat, damage, fire, weather, player health — and no infinite anything. The sea is the antagonist.
- **First build:** a compact dive boat — winch, storage, a cabin, and the range to leave the island chain.
- **House rules:** no fast-travel, no shortcuts; every expedition is an out-and-back with the fuel you carry; recover something from every dive; the submarine must be able to surface on its own ballast.
- **Goal:** chart the whole map, raise the deep wrecks, stand on the arctic ice pack, and bring every crew member home.
- **Mods that shine:** ARCTIC+ (the ice shelf is your frontier), R.A.M.B.O. (ballast and buoyancy physics are the whole game down there), Opal Shader (underwater and night atmosphere), Echo's Bigger Workbenches, Realistic Foghorn.

Watch out

The Salvage Diver is the hard mode of this list — everything is a resource, including air. If an expedition stops being fun, relax one survival toggle; the persona is a lens, not a vow.

4.7 How to start a persona run

1. Choose your layer: Layer 1 (new player) or Layer 2 (fresh save, experience under your belt).
2. New Game → set the mode and survival options for your persona (each persona above lists them).
3. **Enabled Mods** → enable the layer's mods (all of them — the personas differ in goals, not in the list).
4. Write your house rules somewhere visible — three lines on paper beats a forgotten promise.
5. Build the persona's first vehicle, and play its priorities from day one.

Persona	Difficulty	Signature mods
The Mariner	Medium	Opal Shader, R.A.M.B.O., Echo's Bigger Workbenches, Realistic Foghorn, Better Thunder
The Aviator	Medium	Aviation Parts, R.A.M.B.O., Opal Shader, Better Thunder, Realistic Turbocharger

The Industrialist	Medium	Mango Cargo Overhaul, ARCTIC+, Structural Parts, Echo's Bigger Workbenches, Opal Shader
The Engineer	Variable	Everything — R.A.M.B.O., Frustum, Aviation Parts, parts expansions, Opal Shader
The Salvage Diver	Hard	ARCTIC+, R.A.M.B.O., Opal Shader, Echo's Bigger Workbenches, Realistic Foghorn